

2020 AMC 10B Problem 12 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 12. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 12

Problem:

The decimal representation of

$$\frac{1}{20^{20}}$$

consists of a string of zeros after the decimal point, followed by a 9 and then several more digits. How many zeros are in that initial string of zeros after the decimal point?

Answer Choices:

- A. 23
- B. 24
- C. 25
- D. 26
- E. 27

Solution:

We have

$$\frac{1}{20^{20}} = \frac{1}{(10 \cdot 2)^{20}} = \frac{1}{10^{20} \cdot 2^{20}}$$

Now we do some estimation. Notice that $2^{20} = 1024^2$, which means that 2^{20} is a little more than $1000^2 = 1,000,000$. Multiplying it with 10^{20} , we get that the denominator is about

$$\underbrace{100 \dots 0}_{26 \text{ zeros}}$$

Notice that when we divide 1 by an n digit number as long as n is not a power of 10, there are $n - 1$ zeros before the first nonzero digit. This means that when we divide 1 by the 27-digit integer

$$\underbrace{100 \dots 0}_{26 \text{ zeros}}$$

there is (D) 26 zeros in the initial string after the decimal point.

OR

$$\begin{aligned} \left\lceil \log \frac{1}{20^{20}} \right\rceil &= \lceil \log 20^{-20} \rceil \\ &= \lceil -20 \log(20) \rceil \\ &= \lceil -20(\log 10 + \log 2) \rceil \\ &= \lceil -20(1 + 0.301) \rceil \\ &= \lceil -26.02 \rceil \\ &= -26 \\ &= (D) \span style="border: 1px solid black; padding: 0 2px;">26 \end{aligned}$$

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](https://www.amc-math.org/) 

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